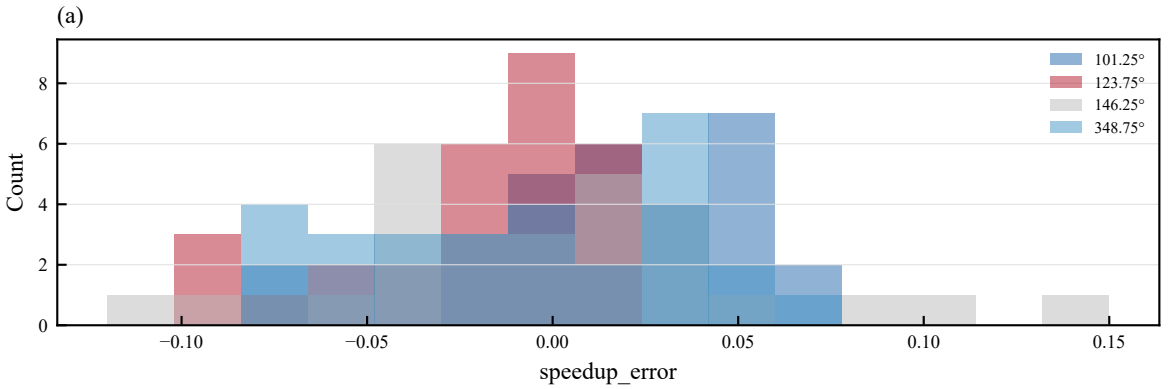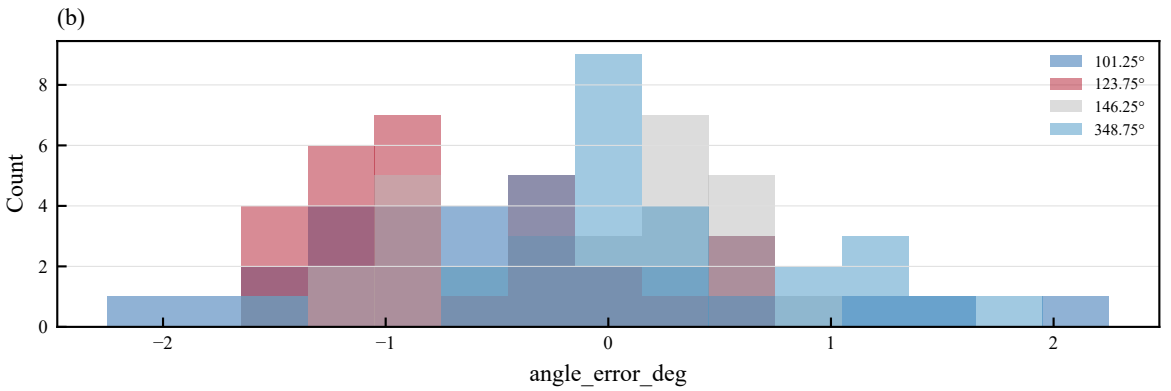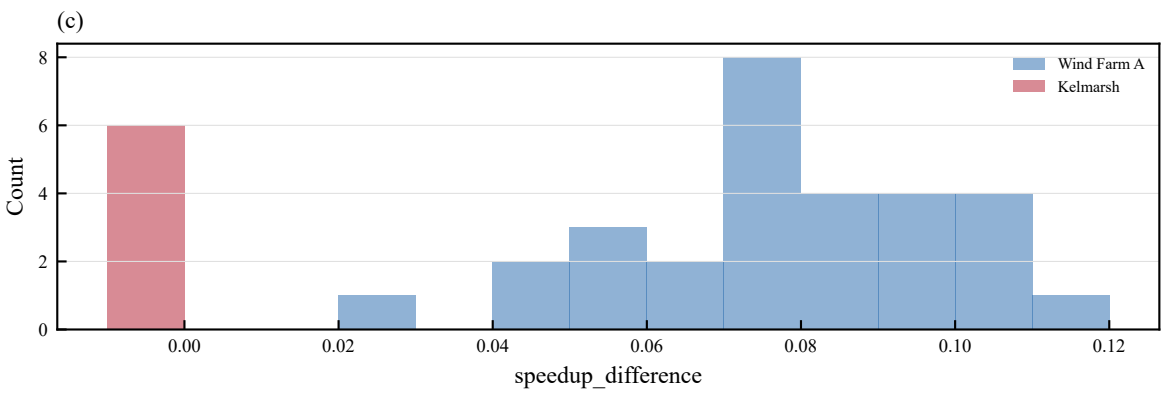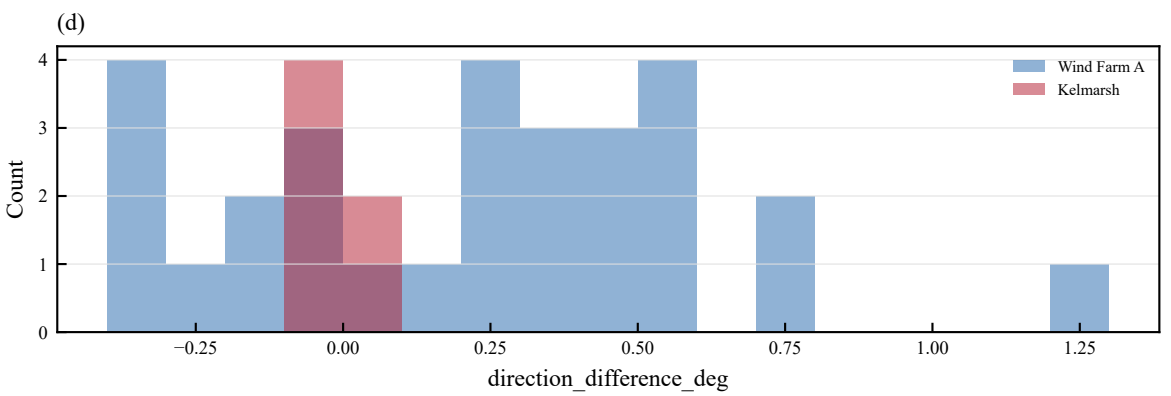CFD quality checks